

EDUCATION**University of Saskatchewan****Saskatoon, SK****4th year of Bachelor of Science Honours, Computer Science (Software Engineering Option)****Cumulative Average: 88%**

SKILLS**Programming Languages:** Expert in Python, Rust, C, C++, SQL, Scala, Prolog, C#, Go, Bash**Software Development:** Agile/Scrum methodologies, UI/UX design, cloud-based solution delivery**Development Tools:** Master in Git (GitHub/GitLab), JIRA, Linux environments, MS Office tools**Machine Learning Frameworks:** PyTorch and TensorFlow Veteran

EXPERIENCES (3+ YEARS)**AI Flood Model Developer – Global Institute for Water Security | Apr–Aug 2026 - Ongoing**

- Research, Design, develop, and train CNN and ConvLSTM models for flood inundation prediction using large-scale geospatial and hydrodynamic datasets, achieving 97% accuracy.
- Built PyTorch-based training pipelines and conducted large-scale experiments on GPUs in HPC systems, reduced error range to just 5cm - 15cm, well below the expected outcome

Full Stack Web & AI Developer Intern – Octyde | Jan 2026 – August 2026

- Developed a full-stack agricultural management system using Svelte and Node.js for real-time farm monitoring, along with an Android app for field operations.
- Debugged 50+ bugs and built interactive geospatial dashboards with MapLibre GL to visualize livestock locations, heatmaps, and farm activity data, improving user satisfaction by 20%.
- Designed and implemented RESTful APIs for device data processing, alarm handling, and real-time status updates, reducing page response time by 50% and improving data refresh latency by 2x.
- Developed a real-time jump-rope counting system using YOLO-based pose estimation with a 20% performance upgrade; applied model quantization (INT8) and deployed on Rockchip RK3588 via RKNN Toolkit, cutting inference time by 40%.

Full-Stack Web & AI developer & scrum master – Roamlil | July 2025 – December 2025

- Utilized modern web technologies, including Python, React, and Next.js to build scalable features and a responsive website, deployed using GCP; Reduced page load times by 35%
- Led and developed the Roamer AI system which integrates LLM with trip planning, achieving 99.9% uptime and smoother user experiences during all test and production environments
- Led Scrum meetings, coordinated task assignments to boost collaboration, conducted testing.

Undergraduate AI Research Intern – University of Saskatchewan | May 2024 – August 2025**Research Focus on Neural network, AI-agents and code clone detection**

- Built AI-agent based bug localization software and led to a publication in IEEE
- Developed real-time AI-assisted code clone detection as VS Code extension, upgrade software quality by reducing 80% code clones
- Invented and implemented a new Convolutional neural network, boost performance by 10%
- Engineered Large Language Model APIs to cut response time by over 50% and delivered efficiency gains for internal tools and workflows across teams

Group Project-Dicom anonymizer – Luxsonic technologies | Jan 2025-April 2025

- Collaborated with a team of 9 to develop a Dicom anonymizer for Luxsonic technologies
- Implement a progressive Web application using React, Typescript, docker.
- Used Jest and Playwright for testing frameworks, achieved 97% test coverage for our system
- Used agile development, developed metrics report using power BI, met stakeholder biweekly

Mobile App developer Intern – All Canadian Grains | September 2024- April 2025

- Collaborated in a team of 5 to develop a weather project app for farmers

- Implemented a responsive mobile interface using React Native and an LLM bot
- Implemented the backend using C # and configured the connection to MySQL, supporting the management of 20,000+ weather and crop-related data records from stations across the globe

Awards/Personal Strengths

- Recipient of entrance scholarship(\$2500), academic excellence award(\$3000)
- Dean's Honor List for 2023-2024 and Member of Golden Key International Honour Society
- Fast learner, always willing to learn, continuously seeking opportunities to learn new skills
- Team player/Leader, led multiple software projects resulting in publications or production
- Language: English: Fluent (written and spoken) Chinese: Fluent (written and spoken)